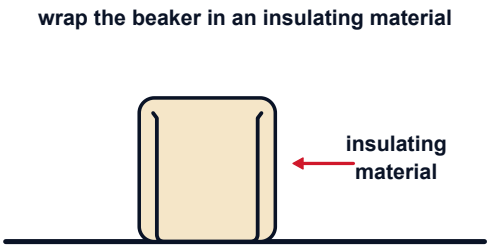
DO NOW · REQUIRED PRACTICAL

Thermal insulation

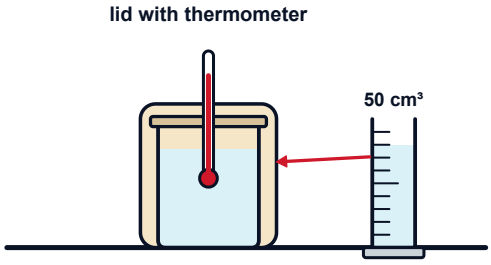
Energy
Paper 1 · Separate Physics only
6 steps

A student compares how well different materials insulate a beaker of hot water.

1 Wrap a beaker in one of the insulating materials.

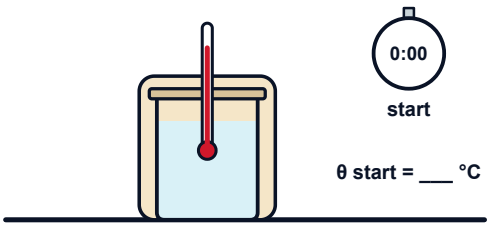


2 Pour in a measured volume of hot water and fit a lid with a thermometer.

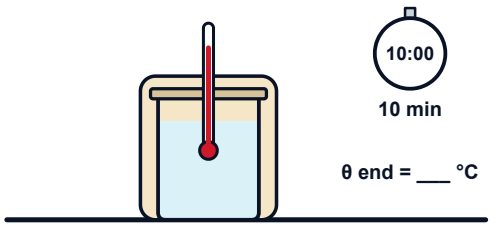


same measured volume of hot water each time

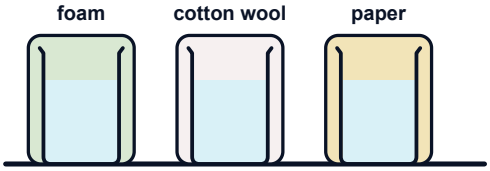
3 Record the starting temperature and start a stopwatch.



4 Record the temperature again after 10 minutes.



5 Repeat with each material, using the same volume and starting temperature of water.



same volume, same starting temperature

6 Compare the temperature falls: the smallest fall shows the best insulator.

material	start	end	fall
foam			
cotton wool			
paper			

fall = start − end

smallest fall = best insulator

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